Week 3: Magnetism, Biot-Savart Law, Ampere's Law, Electromagnetic Induction, Maxwell's Equations and Magnetic Materials SOLUTIONS

Discussion Questions

1 a Use $\boldsymbol{F} = q\boldsymbol{v} \times \boldsymbol{B}$ or a right hand rule

2 b and **d** Magnetic field force does no work (perpendicular to motion), but clearly velocity and momentum directions change, as does the acceleration direction (towards the centre of the circular path)

3 e Use right hand grip or corkscrew rule

4 First work out the direction of the magnetic field (right hand grip rule), then use $\boldsymbol{F} = q\boldsymbol{v} \times \boldsymbol{B}$ or a right hand rule *OR* observe that parallel currents always attract.

5 Use Lenz's Law – field is towards the eye and increasing, induced current will generate field in opposite direction, i.e. clockwise.

6a Field is increasing into page, induced current will generate field out of the page (current is anticlockwise) **b** same answer **c** opposite reasoning to 2a and 2b (current is clockwise)

7i Gauss' Law for magnetism has non-zero RHS corresponding to strength of monopoles enclosed by Gaussian surface. **ii** extra term in Faraday's Law corresponding to flow of magnetic monopoles (see lecture notes for excellent coverage)

Problem Solving Questions

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a) $v_{\parallel} = v\cos 30° = 1.3\times10^{7}\text{m/s} \times \cos 30° = \mathbf{1.1\times10^{7}\,m/s}$.

Rate of change of $v_{\parallel}$ is zero because the force along the magnetic field is zero.

b) $v_{\perp} = v\sin 30° = 1.3\times10^{7}\text{m/s} \times \sin 30° = \mathbf{6.5\times10^{6}\,m/s}$.

c) The magnetic force is perpendicular to the velocity. Hence the magnetic force changes the direction of the velocity, not its magnitude, thus producing centripetal acceleration. This is true at every instant, so the velocity vector rotates about the field.

d) The electron traces out a helical path whose axis lies parallel to the magnetic field.

e) Force = mass × centripetal acceleration. Mathematically, $qv_{\perp}B = \frac{mv^2}{R}$, so $R = \frac{mv_{\perp}}{qB} = 9.1\times10^{-31}\text{kg}\times6.5\times10^{6}\text{ms}^{-1}/\,(1.6\times10^{-19}\text{C}\times0.2\text{ T}) = \mathbf{1.85\times10^{-4}\,m}$

f) $T = \frac{2\pi R}{v_{\perp}} = 2\pi\times1.85\times10^{-4}\text{m}/\,6.5\times10^{6}\text{ms}^{-1} = \mathbf{1.79\times10^{-10}\,s}$

g) Distance along field during one period = $v_{\perp}T = 1.13\times10^{7}\text{ms}^{-1}\times1.79\times10^{-10}\text{s} = \mathbf{2\times10^{-3}\,m}$

9

(a) The ions which exit through the slit are those which move from the source to the slit along a horizontal line. The net force on these ions perpendicular to their straight-line path must be zero, otherwise they would be deflected. So we have the condition $qvB = qE$, which is only satisfied by ions with $v = E/B$. Ions with other speeds are deflected up or downward (parallel or anti-parallel to $\boldsymbol{E}$) according to whether $v < E/B$ or $v > E/B$ respectively.

(b) Ions are deflected downward along a semicircular path (right hand rule interaction). From the relation $R = mv/qB$, the element with greatest mass describes the path with the largest radius. Given that the unknown element strikes the plate between the two others, it cannot be the one. Of the other two, oxygen is more massive (m = 16u, vs carbon's m = 12u).

(c) $R_O = \frac{m_O v}{qB}$, $R_C = \frac{m_C v}{qB}$ and $R_x = \frac{m_x v}{qB}$, where the subscripts O, C and x denote oxygen, carbon and the unknown element respectively. Then $2(R_O - R_C) = 2.25$ cm and $2(R_x - R_C) = 1.16$ cm. Taking the ratio of these: $(R_O - R_C)/(R_x - R_C) = (m_o - m_C)/(m_x - m_C) = 2.25/1.16$. Solving for m_x gives $\mathbf{14}\boldsymbol{u}$, which corresponds to nitrogen.

10

a) Positive charges move to the right and negative charges move to the left (by current conventions).

b) Upper side is positive with respect to the lower one, since its potential is higher.

c) Positive. A positive charge moves to the right to create the current *i*. The right hand rule describes the direction of the magnetic force it feels to be upwards (because the magnetic field is directed into the page). The particle carrying the charge therefore migrates to the upper side, charging the point S positively. To make sure you understand what is going on, think about what would happen if the charge carriers were negative instead. They would move to the left to create the conventional current *i*. The right hand rule again predicts that they would feel a magnetic force pointing *up* again! [Remember: negative charges reverse the right hand rule, following a *left hand rule*.] They would therefore migrate to the upper side, charging the point S negatively, so that it would be at a *lower* potential.

11a Use $R = mv/qB$; answer is **0.224 m**. **11b** $f = \frac{v}{2\pi R} = 0.1\times3\times10^8\text{ms}^{-1}/\ 2\pi\times0.224\text{ m} =$ $\mathbf{2.13\times10^7\,Hz}$. The oscillator frequency should match the orbital period of the electrons in the cyclotron so that they are accelerated as efficiently as possible (like pushing a child "in sync" on a swing).

12

(a) The direction will be perpendicular to the plane formed by the two long wires; the contribution from the 12 A wire will be greater than that from the 8 A wire, and in the opposite direction (use right hand grip rule).

(b) Use $B = \frac{\mu_0(I_1-I_2)}{2\pi d}$ to give $\mathbf{4.0\times10^{-4}\ T}$

(c) Use inverse proportion to give 0.24 cm from the 12 A wire and hence **0.16 cm** from the other

13 Integrate $dB = \frac{\mu_0 i dl\times\hat{r}}{4\pi r^2}$ maround the semicircle; it's simple because the cross product simplifies to $\sin\theta = \sin 90° = 1$, and r is constant. Answer is hence $B = \frac{\mu_0 i\pi r}{4\pi r^2} = \boldsymbol{\frac{\mu_0 i}{4r}}$.

14 Use $B = \mu_0 ni$ with n = 1/0.0008 and i = 2.0; answer is $\mathbf{3.14\times10^{-3}\ T}$.

15 Use Ampère's Law; use circles centred on the axis of the cylinders.

16b Use a rectangular Ampèrean loop symmetrically located either side of the sheet.

17 Key formula: emf (average) = $n\cdot\Delta\phi/\Delta t$; also use $\phi = BA$; **a)** doubling field doubles flux and hence flux change = original flux; 12.6 V **b)** reducing field to zero is same change as part (a), hence 12.6 V, **c)** reversing field means flux change = double flux; hence 25.1 V **d**)the same as **b**; **e)** the same as **c**

18i and **ii** track flux change, use Lenz's Law, **iii** change in flux = $0.4\cdot\pi(.05)^2$; n = 100, Δt = 0.005; use emf = $n\cdot\Delta\phi/\Delta t$; answer is 63 V

19 Find magnitudes of magnetic fields at wire 4 from other three wires, using Ampèrean loops.

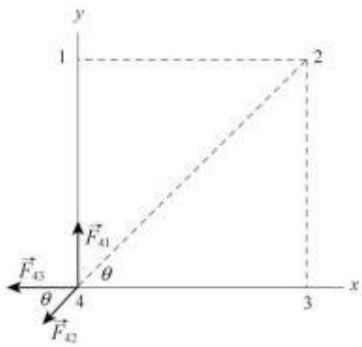
$$|B_{41}| = |B_{43}| = \frac{\mu_0 i}{2\pi a} \;\&\; |B_{42}| = \frac{\mu_0 i}{2\sqrt{2}\pi a}$$

Force experienced by wire #4 given by the Lorentz force for currents:

$F = il \times B = ilB\sin\theta$, where $\theta = 90° \therefore \sin 90 = 1$

$\therefore F = ilB \rightarrow \frac{F}{l} = iB$ (force per unit length)

$$\left|\frac{F_{41}}{l}\right| = \left|\frac{F_{43}}{l}\right| = \frac{\mu_0 i^2}{2\pi a} \;\&\; \left|\frac{F_{42}}{l}\right| = \frac{\mu_0 i^2}{2\sqrt{2}\pi a}$$



From geometry of forces, find components in x and y:

$$\frac{F_{Net}}{l}|_x = -\frac{F_{43}}{l} - \frac{F_{42}}{l}\cos 45° = -\frac{\mu_0 i^2}{2\pi a} - \frac{\mu_0 i^2}{4\pi a} = -\frac{3\mu_0 i^2}{4\pi a}$$

$$\frac{F_{Net}}{l}|_y = \frac{F_{41}}{l} - \frac{F_{42}}{l}\sin 45° = \frac{\mu_0 i^2}{2\pi a} - \frac{\mu_0 i^2}{4\pi a} = \frac{\mu_0 i^2}{4\pi a}$$

$$\therefore \frac{F_{Net}}{l} = \left(-\frac{3\mu_0 i^2}{4\pi a}, \frac{\mu_0 i^2}{4\pi a}\right) \text{N/m}$$

In (i,j) unit vector notation, substituting the values for i = 7.5 A and a = 13.5 cm:

$$\frac{F_{net}}{l} = (-1.25\times10^{-4}\text{Nm}^{-1})\hat{\boldsymbol{\imath}} + (4.17\times10^{-5}\text{Nm}^{-1})\hat{\boldsymbol{\jmath}}$$

20 **a b** see short answers **c** viewed from above, flux is increasing upwards, induced current will produce flux downwards (clockwise)

21 $i_d = \varepsilon_0 \frac{d\Phi_E}{dt} = \varepsilon_0 \frac{d}{dt}(AE) = \varepsilon_0 A \frac{d}{dt}\left(\frac{V}{d}\right) = \frac{\varepsilon_0 A}{d}\left(\frac{dV}{dt}\right)$ OR $\frac{dV}{dt} = \frac{i_d d}{\varepsilon_0 A} = \frac{i_d}{C} = \frac{1.5\text{A}}{2.0\times10^{-6}\text{F}} = 7.5\times10^5 \frac{\mathbf{V}}{\mathbf{s}}$. Therefore we need to change the voltage difference across the capacitor at the rate of 7.5×10^5V/s.

Past Exam Questions

22i Use $\frac{mv^2}{r} = qvB$ to give **2.6 ×10⁶ m/s** **ii** Use $f = \frac{v}{2\pi r}$ to give **9.2 MHz**

23 a & b Use right hand grip rule to give +x; then use right hand slap (or other) rule to give –y.

24i The Biot-Savart Law $dB = \frac{\mu_0 i dl \times \hat{r}}{4\pi r^2}$ gives the field at each point, where $r = \sqrt{R^2 + z^2}$.
Now find the axial component (all the others cancel, by symmetry). This gives:

$$B = \frac{\mu_0 i 2\pi R^2}{4\pi(z^2 + R^2)^{\frac{3}{2}}}$$

ii As z gets very large, denominator changes $z^2 + R^2 \rightarrow z^2$, and this simplifies to the required expression.

25 Shape an answer from the text (HRW) or from *Hyperphysics.* **Use dot points** and as many diagrams as you can manage.

26a At 1: zero; 2: CW; 3: zero; 4: ACW; 5: zero; 6: CW; 7: zero **bi** away **bii** away (imagine spheres as bar magnets with magnetic moments pointing to the left) **biii** less than (field of bar magnet less at the greater distance)